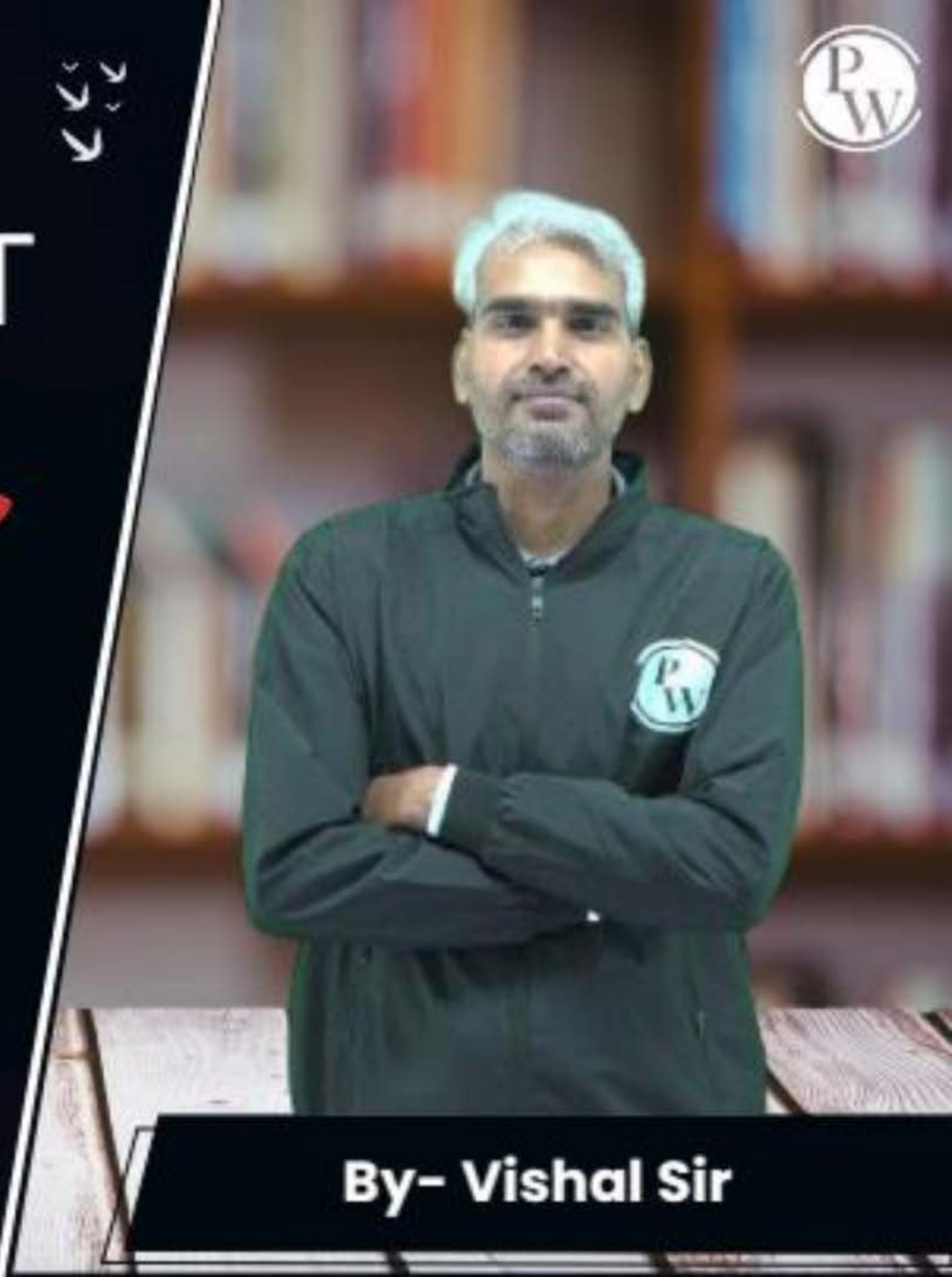
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- Topic: Vertex covering
- Topic: Vertex independent set
- Topic: Connectivity

Slide

Topics to be Covered



Topic Connectivity

Topic Spanning tree

[redacted]

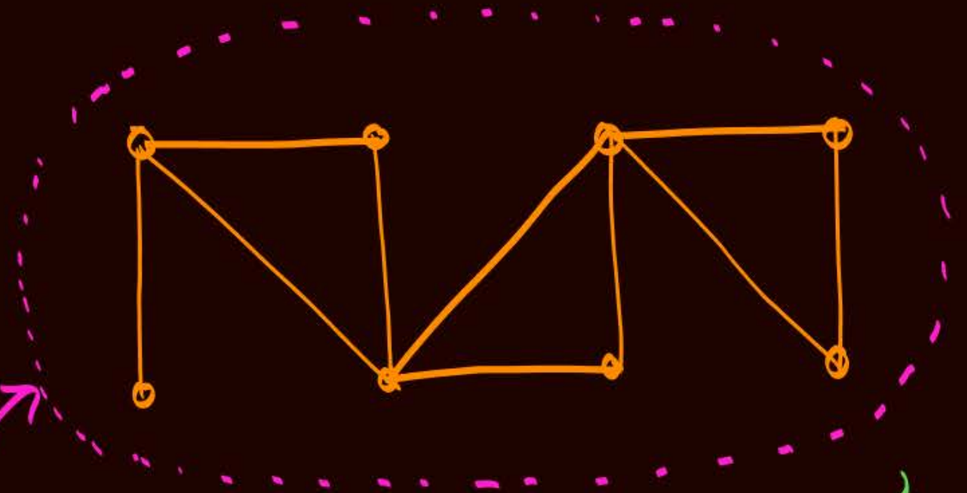
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Topic : Connectivity

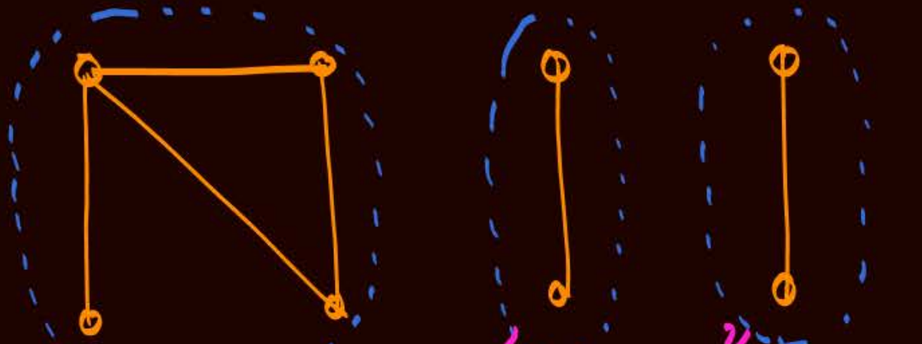
A graph is said to be connected if there exists a path between every pair of vertices.

If a graph (with 2 or more vertices) is not connected, then it will have two or more connected components.



Single Connected Component

Connected graph



Three Connected Components

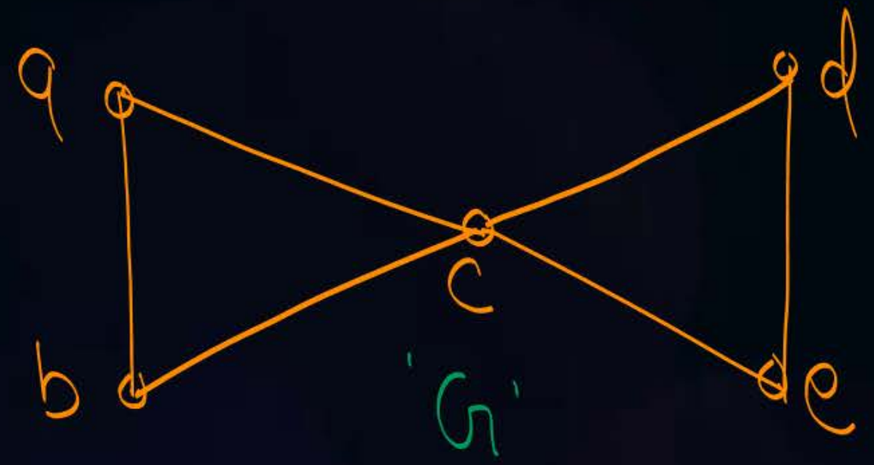
disconnected graph



Topic : Cut-vertex

"Articulation point"

Let G be a connected graph, a vertex $v \in G$ is called a cut vertex if deletion of vertex "v" from graph G makes the graph disconnected.



it is a Connected graph

delete 'c'



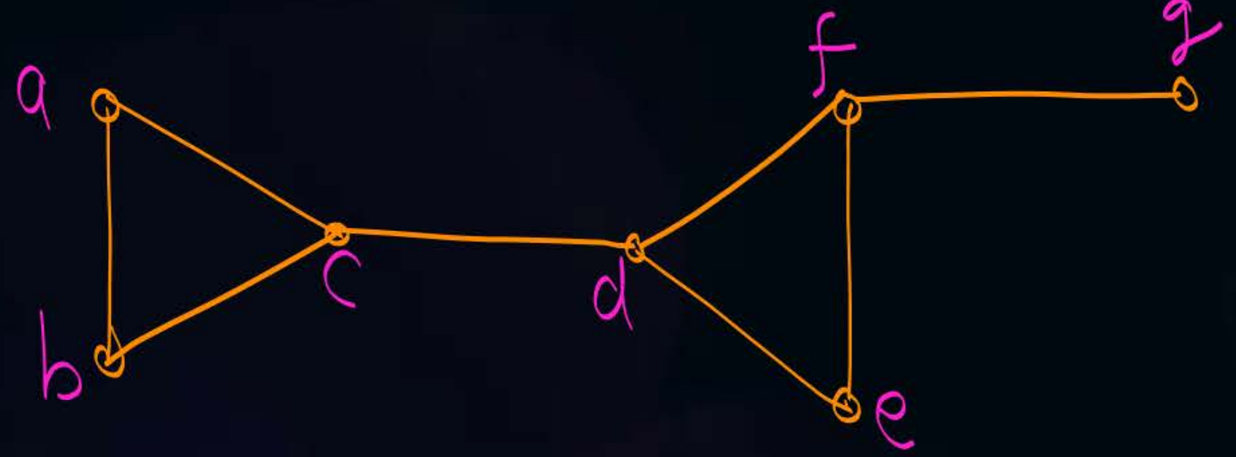
Graph becomes disconnected

∴ 'c' is a Cut-vertex



Topic : Cut-edge / Bridge

Let G be a connected graph, an edge $e \in G$ is called a cut edge or bridge if deletion of edge "e" from graph G makes the graph disconnected.



Connected graph

Edge $\{c,d\}$ is a Cut edge as well as Edge $\{f,g\}$ is also a Cut edge



Topic : NOTE

① An edge of a connected graph G can be a cut-edge (bridge) if and only if that edge is not a part of any cycle in the graph.

② At least one vertex associated with a cut-edge is a cut-vertex of that graph

eg: w.r.t. cut-edge {c,d} in above example, both c & d are cut-vertex

w.r.t. cut-edge {f,g} in above example, 'g' is not a cut-vertex but 'f' is a cut-vertex

Note:- Existance of Cut-vertex in a graph does not imply the existance of Cut-edge, but

Existance of a Cut-edge in a Graph implies the existance of Cut-vertex.



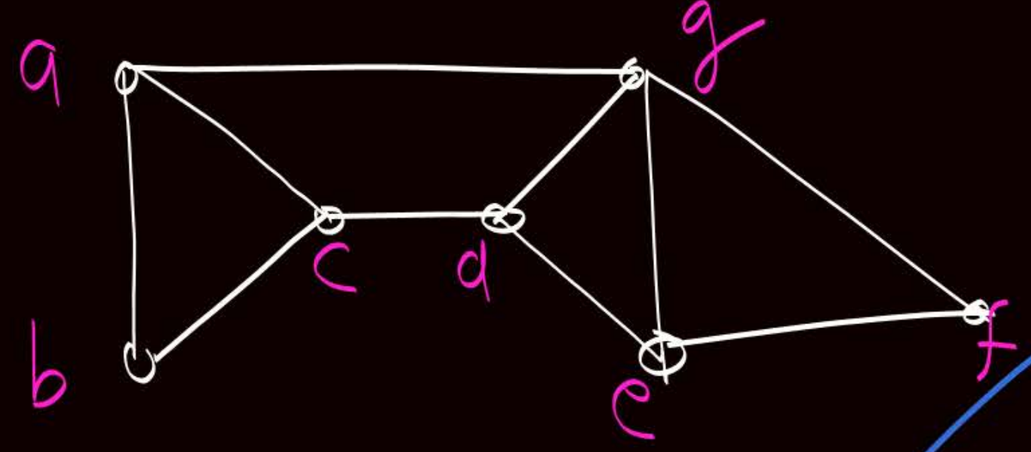
Topic : Cut-set

"No edge can be removed"

A minimal set of edges in graph G, such that removal of all the edges of that set from the graph G makes the graph disconnected is called cut-set of graph G.

& connected

If we leave any edge of that set and delete the remaining edges then it should never disconnect the Connected.



The set of edges
$\{\{c,d\},\{d,e\},\{a,g\}\}$ is
a Cut set or not?

Removal of all the edges
can make the graph
disconnected, but it is
not a Cutset, because it
is not minimal
(We can remove the edge $\{d,e\}$
from the set

Here $\{\{c,a\},\{a,g\}\}$ is
a Cut-set.



Topic : Vertex connectivity

- Let G be a connected graph, the minimum number of vertices whose removal from graph G makes the graph disconnected is called vertex connectivity of graph G.
Vertex connectivity of a graph G is denoted by $\kappa(G)$

- Vertex connectivity of graph is 1 if and only if cut-vertex exists in the graph.

- If graph G is disconnected graph, then $K(G) = 0$



Topic : Edge connectivity

Let G be a connected graph, the minimum number of edges whose removal from graph G makes the graph disconnected is called edge connectivity of graph G.

Edge connectivity of a graph G is denoted by $\lambda(G)$

- Edge connectivity of graph is 1 if and only if cut-edge exists in the graph.
- If Graph G is a disconnected graph, then $\lambda(G) = 0$

Note:-

① For a connected graph G,

$$\lambda(G) \leq \delta(G) \quad \text{---eq}^{n}\ (1)$$

② For a connected graph G

$$K(G) \leq \lambda(G) \quad \text{---eq}^{n}\ (2)$$

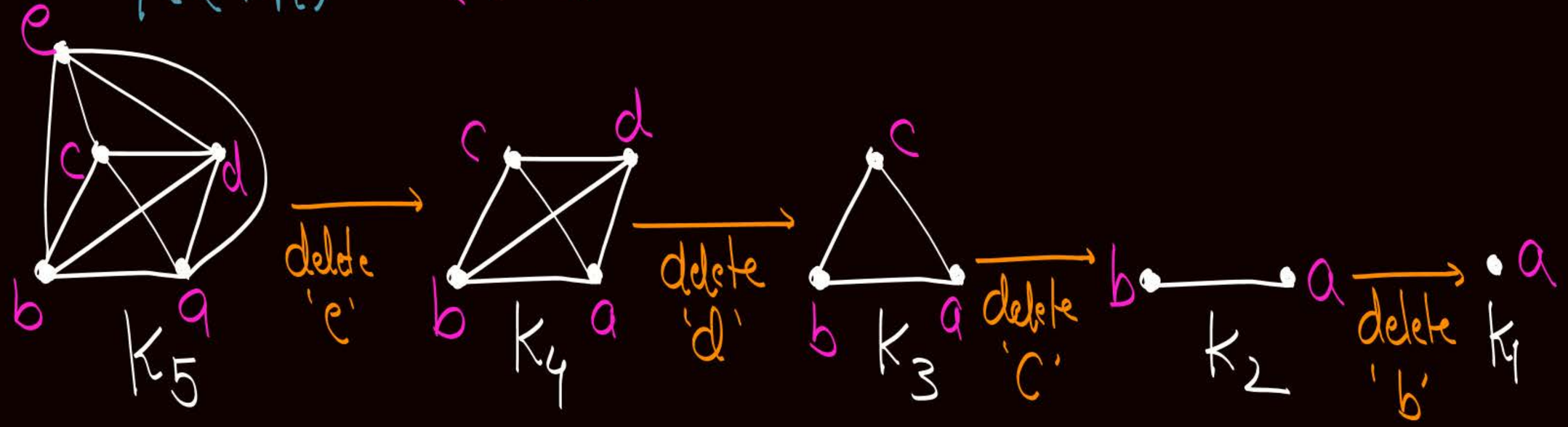
By eq^{n} ① & ②

$$K(G) \leq \lambda(G) \leq \delta(G)$$

Today's Topic

Q:- Find $\lambda(G)$ and $K(G)$ for Complete graph K_n

$$\lambda(K_n) = (n-1)$$



$$K(K_n) = (n-1)$$

Note: If an isolated vertex exists along with other components in the graph, then isolated vertex is treated as a connected component

- If there is only '1' isolated vertex, and nothing else, then it is exceptional case in which it may be treated as connected or disconnected as per situation

H.W: Find Vertex Connectivity ($K(G)$) and Edge Connectivity ($\lambda(G)$) For the following graphs.

	Vertex Connectivity	Edge Connectivity
① Complete graph K_n	$(n-1)$	$(n-1)$
② Cycle graph C_n	2	2
③ Wheel graph W_n	$1 + 2 = 3$ (1: Hub, 2: wrt "C_{n-1}")	3
④ Complete bipartite graph $K_{m,n}$	$\text{Min}(m,n)$	$\text{Min}(m,n)$
⑤ Star graph with n-vertices	1	1

H.W: {already done}

#Q. If G is a forest with n vertices and k connected components, how many edges does G have?

Every Component is a tree.

$$Ans = (n - k)$$

H.W. [MCQ]

#Q. In a connected graph G if we delete an edge then no. of components are

A. 0 or 1

B. 1 or 2

C. 2

D. ≥2

1 = w.r.t. edge being deleted is not a cut-edge

2 = When edge being deleted is a cut-edge

[MCQ]

HW

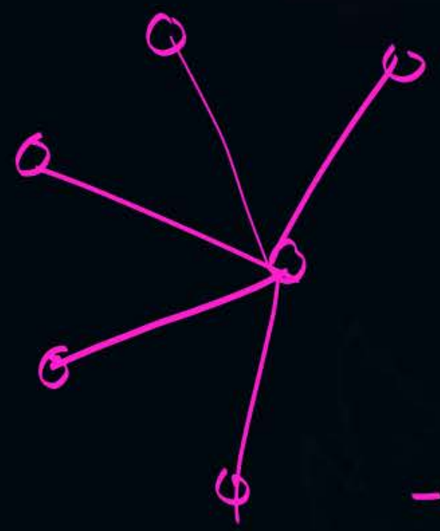
#Q. In a connected graph G with 'n' vertices ($n \geq 2$) if we delete a vertex then number of components are

A. ≤ 2

B. 1 or 2

C. Lies between 1 and n -1

D. ≥ 2



Star graph with 'n' vertices
(delete hub vertex)

Q: Let G be a graph with 'n' vertices of 'K' Components, if we delete a vertex from graph G then no. of Components can lie in the range from (K-1) to (n-1)

(K-1): if vertex being deleted is an isolated vertex

H.W.

#Q. A simple graph with n vertices is necessarily connected if number of edges are more than E, then find the value of E.

Isolated vertex

Complete graph with $(n-1)$ vertices and $(n-1)C_2$ edges

It is a disconnected graph with 'n' vertices & $(n-1)C_2$ edges

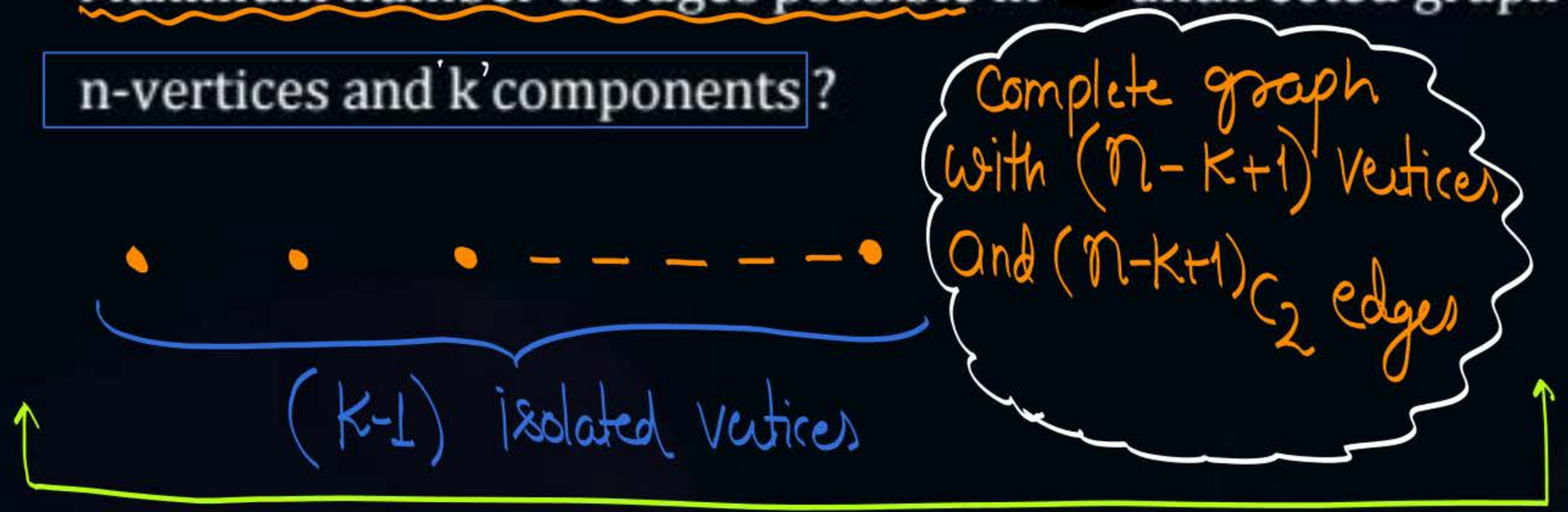
With upto $(n-1)C_2$ edges a graph with 'n' vertices may be disconnected.

* But if no. of Edges are more than $(n-1)C_2$ then simple graph with 'n' vertices is Necessarily Connected

$\therefore E = (n-1)C_2 = \frac{(n-1)(n-2)}{2}$

H.W.

#Q. Maximum number of edges possible in a simple undirected graph with n-vertices and 'k' components?

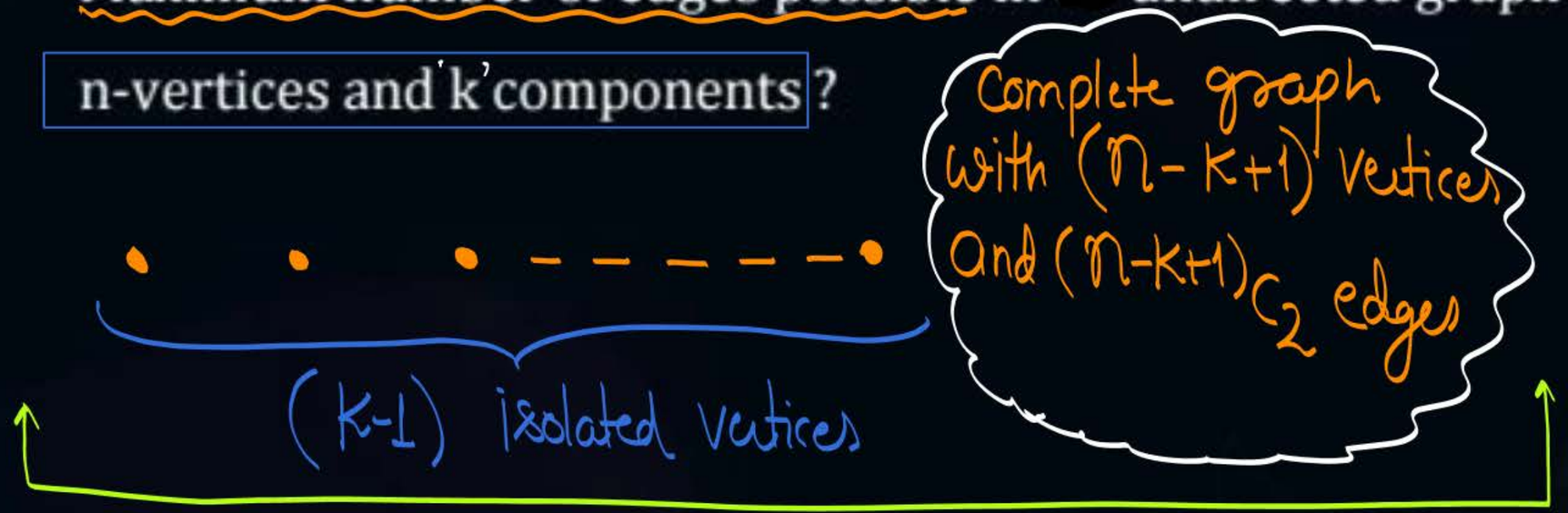


It is a graph with 'n' vertices, 'k' Components and $(n-k+1)C_2$ edges.

* Upto $(n-k+1)C_2$ edges it may be possible to have a graph with 'n' vertices and 'k' Components, but if no. of Edges are more than $(n-k+1)C_2$ then no. of Components can never be 'k' {it will be less than k}

H.W.

#Q. Maximum number of edges possible in a simple undirected graph with n-vertices and 'k' components?



It is a graph with 'n' vertices, 'k' Components and $(n-k+1)C_2$ edges.

Maximum number of Edges Possible in a Simple graph with 'n' vertices & 'k' Components $= (n-k+1)C_2 = \frac{(n-k+1)(n-k)}{2}$

Note: A simple graph with 'n' vertices & 'k' components must have at least $(n-k)$ edges and can have at most $(n-k+1)C_2$ edges.

Spanning Tree



Topic : Spanning Tree

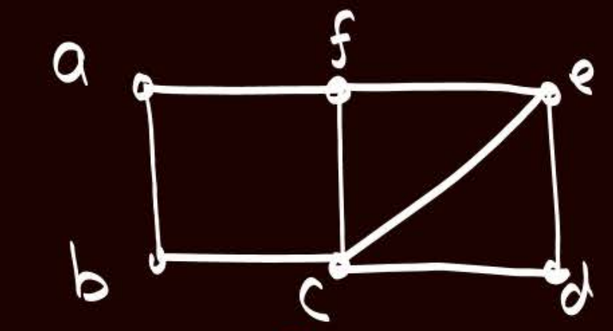
Let G be a connected graph,

If graph is not a Connected graph, then spanning tree does not exist for that graph.

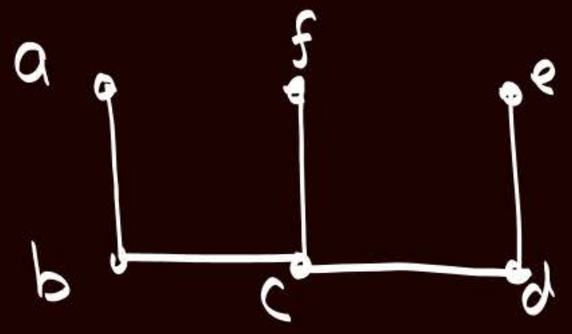
- A sub-graph H of graph G is called a spanning tree if and only if
  - H is a tree (Acyclic + Connected)
  - and H contains all vertices of graph G

A graph G is connected if and only if Spanning tree exist for graph G

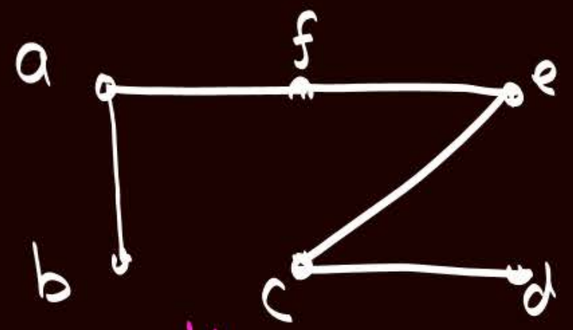
eg



G_1

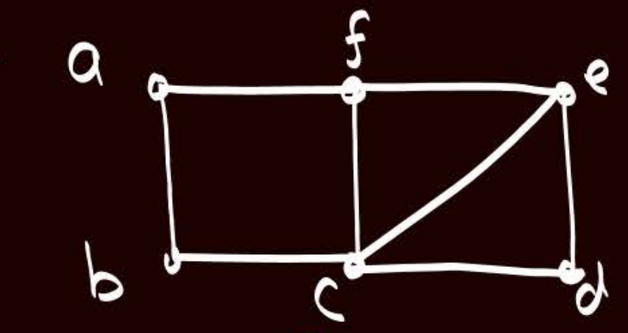


H_1

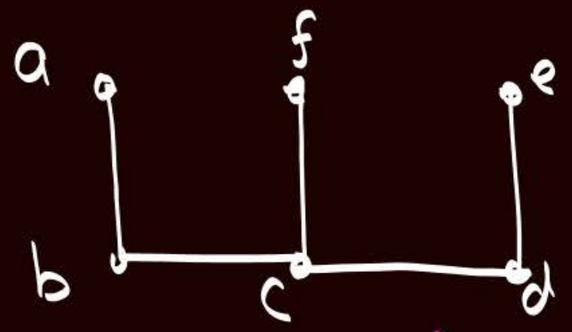


H_2

eg



G

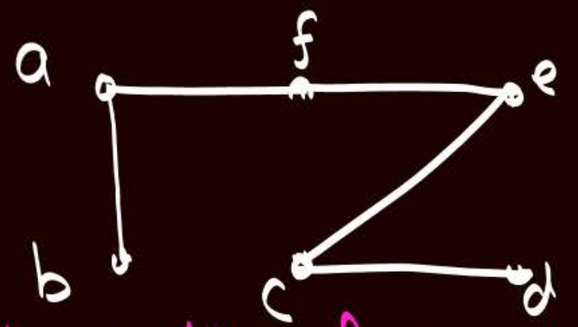


Sub-graph H_1 of graph G

Acyclic ✓ + Connected ✓ + Contain all vertices ✓

tree

∴ H_1 is a spanning tree of G

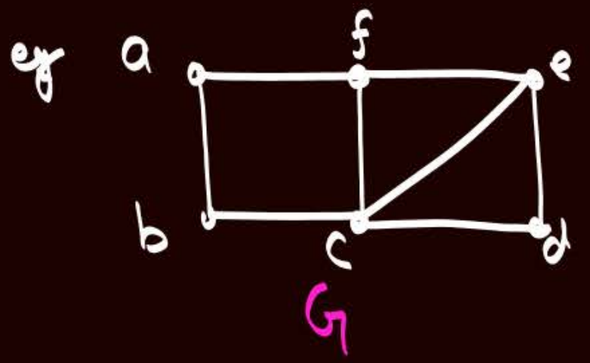


Sub-graph H_2 of graph G

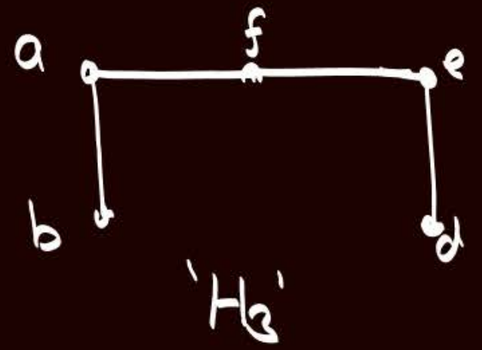
Acyclic ✓ + Connected ✓ + Contain all vertices ✓

∴ H_2 is a spanning tree of G

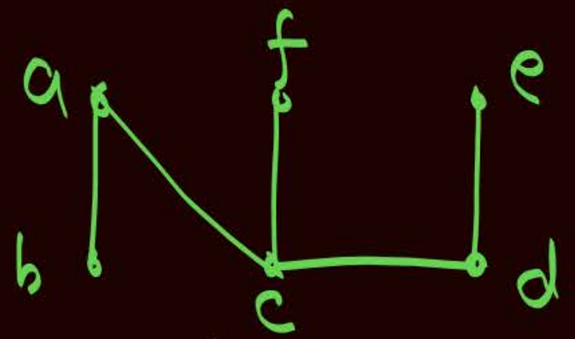
For a connect graph G,
Spanning tree of G
need not be unique



G

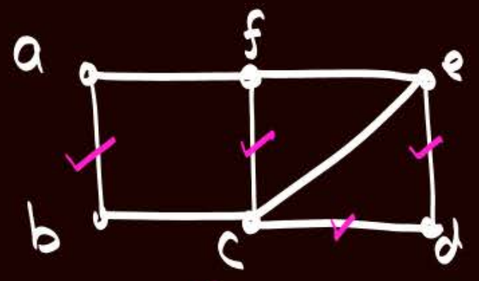


'H3'



H4

eg

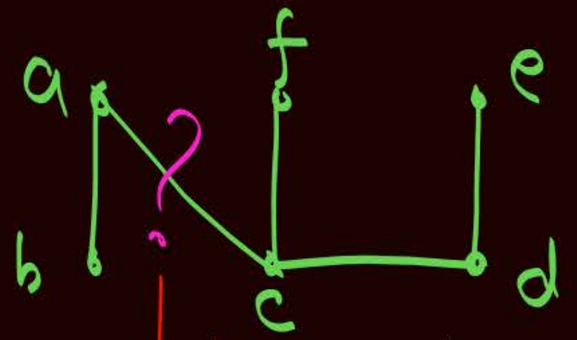


G



'H_3'
- it is a subgraph of G ✓
- it is a tree ✓
- It is acyclic ✓
- It is connected ✓

But it does not contain all vertices of graph G. ∴ it is not a Spanning tree of G.



H_4
- it is a tree ✓
- It is acyclic ✓
- It is connected ✓
- Contains all vertices of G ✓

edge $\{a,c\} \notin G$, ∴ H_4 is not a subgraph of G. Hence, H_4 is not a Spanning tree of G.

② If graph G is connected it will have at least '1' spanning tree

③ Graph G will have exactly '1' spanning tree if and only if graph G itself is a tree.

{ in that case, spanning tree will be graph G itself }

Note: Let G be a connect graph with n-vertices, then each spanning tree of G will contain exactly (n-1) edges.



Topic : Circuit Rank

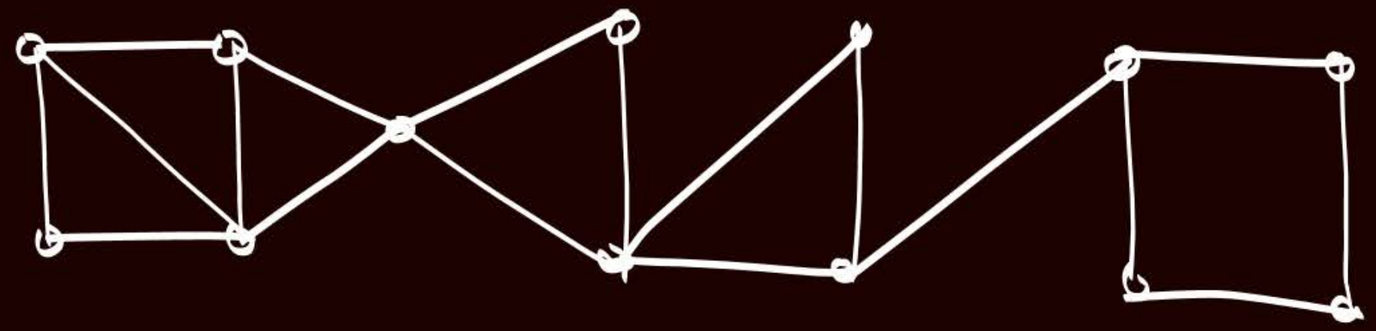
- Number of edges that must be deleted from a connected graph G in order to obtain a spanning tree of G is called circuit rank of graph G.

* Let G_1 is a Connected graph with n-vertices and m-edges.

then Circuit Rank of graph $G_1 = m-(n-1) = (m-n+1)$

eg Find the Circuit rank of the following graph.

G =

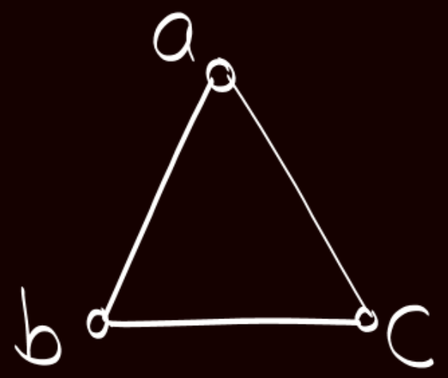


Vertices in G = 13

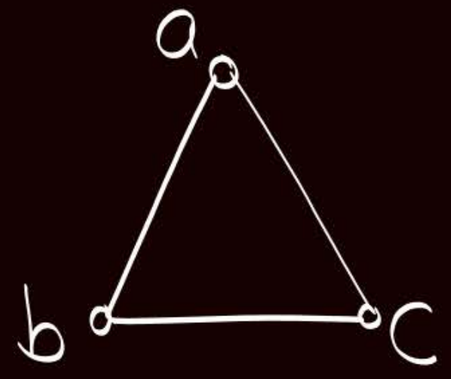
Edges in G = 18

Circuit Rank of G = $18-(13-1)$ = 6

Q:- Find the no. of spanning trees of following graph



Q:- Find the no. of spanning trees of following graph



Circuit Rank = 3 − 2 = 1

There is a cycle of length = 3

→ From a cycle of length = 3
We can delete an edge in $^3C_1$ ways

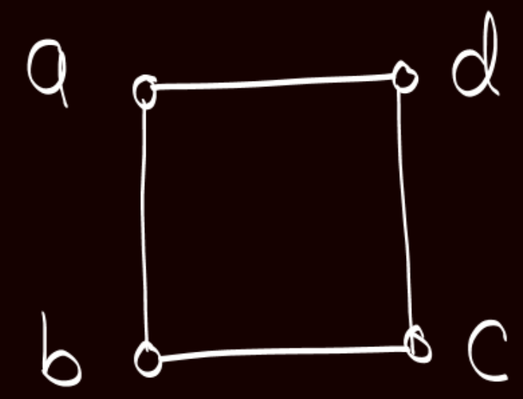
and

* Hence '3' spanning trees are possible



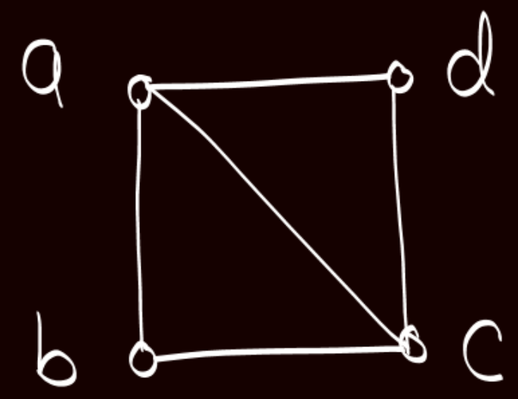
'3' spanning tree

Q:- Find the no. of spanning trees of following graph

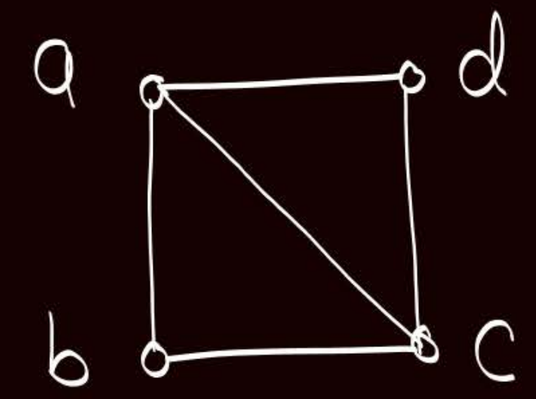


Ans = 4

Q:- Find the no. of spanning trees of following graph



Q:- Find the no. of spanning trees of following graph



No. of vertices = 4

No. of Edges = 5

Circuit Rank = $5 - 3 = 2$

Out of '5' edges we need to delete '2' edges

Total no. of spanning trees = Total no. of ways to delete '2' out of '5' edges — No. of ways to delete '2' edges such that resulting subgraph is not a tree

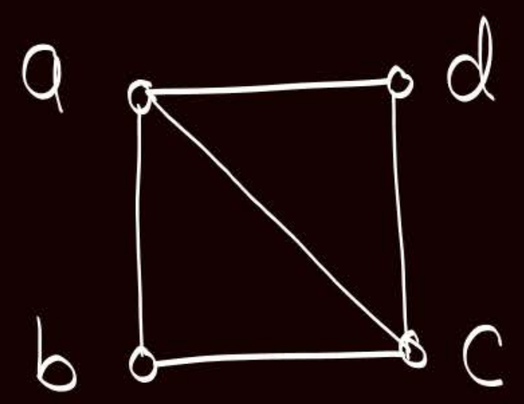
$= {}^5C_2 - 2$

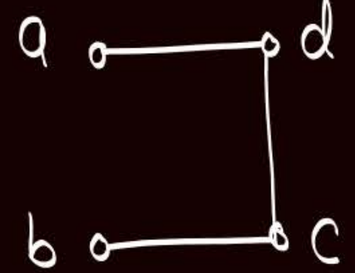
$= 10 - 2$

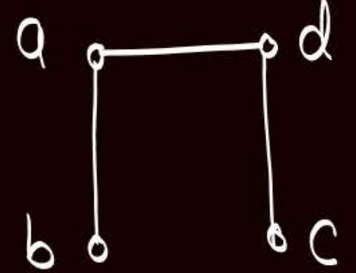
$= 8$

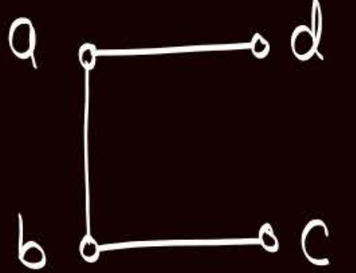
{ w.r.t edges $\{a,b\}, \{b,c\}$
$\equiv$
w.r.t. edges $\{a,d\}, \{c,d\}$

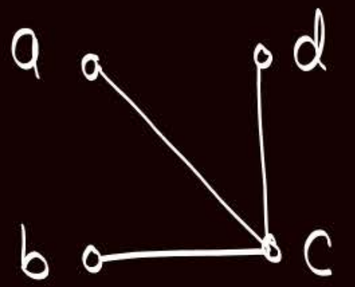
Q:- Find the no. of spanning trees of following graph

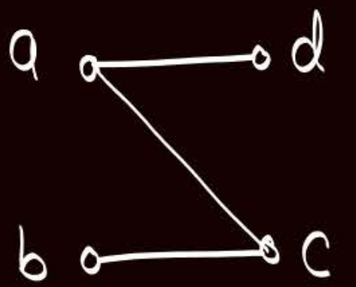


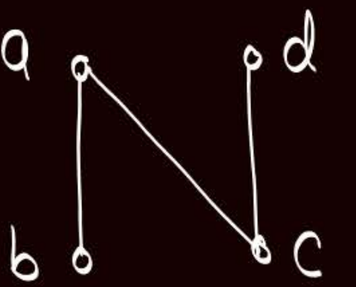


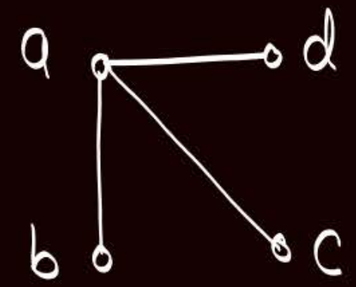






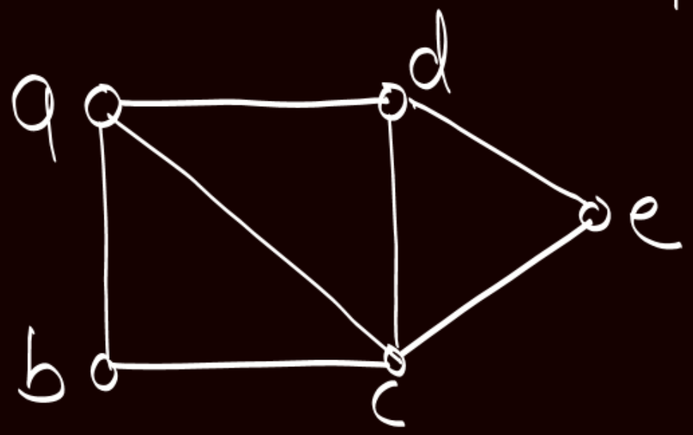






HW

Q:- Find the no. of spanning trees of following graph



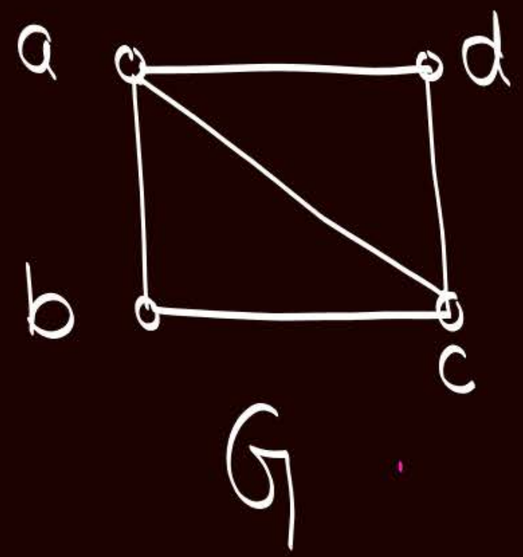
✯ Kirchoff's Matrix Theorem :-

Kirchoff's matrix theorem can be used to find the number of spanning trees of a given graph.

★ Kirchoff's Matrix Theorem:-

Adjacency matrix is a way of representing the graph

Step-1 :- Let 'A' is the adjacency matrix w.r.t. given graph



G

⟹ Adjacency matrix w.r.t. graph G

$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

* Kirchoff's Matrix Theorem:-

Step-2 • 'M' is a matrix obtained from adjacency matrix 'A' by replacing all 1's of matrix A by '-1' and by replacing all the 0's of principle diagonal of matrix 'A' by the degrees of corresponding vertex.

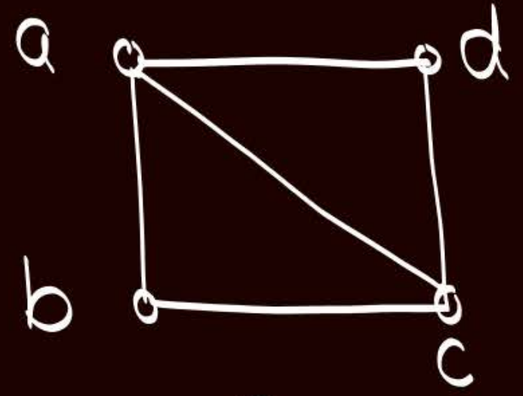
{ All other 0's of matrix 'A' will remain as it is }

$$M = D - A$$

D → Degree Matrix w.r.t. given graph

Matrix M is called Laplacian Matrix

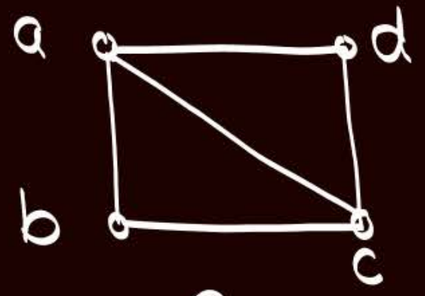
✯ Kirchoff's Matrix Theorem:-



G

$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array} \Longrightarrow M = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{array}$$

✯ Kirchoff's Matrix Theorem:-



G

$\Rightarrow$ Degree matrix w.r.t. graph G: $D =$

	a	b	c	d
a	3	0	0	0
b	0	2	0	0
c	0	0	3	0
d	0	0	0	2

$M = D - A$

$M =$

	a	b	c	d
a	3	0	0	0
b	0	2	0	0
c	0	0	3	0
d	0	0	0	2

$-$

	a	b	c	d
a	0	1	1	1
b	1	0	1	0
c	1	1	0	1
d	1	0	1	0

$=$

	a	b	c	d
a	3	-1	-1	-1
b	-1	2	-1	0
c	-1	-1	3	-1
d	-1	0	-1	2

- Laplacian matrix is a Symmetric matrix
- In laplacian matrix, summation of the elements of each row and each column is always "Zero"

  ∴ Laplacian matrix is a Singular matrix

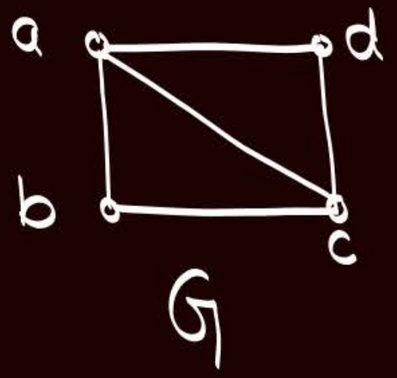
  i.e. determinant value = 0

✱ Kirchoff's Matrix Theorem:-

- Once we have obtained matrix M, Co-factor w.r.t. any element of matrix M will give us the number of spanning trees corresponding to the given graph

Co-factor of $a_{ij} = (-1)^{i+j} *$ determinant value while skipping elements of ith row & j^{th} Column.

★ Kirchoff's Matrix Theorem:-



$$\Rightarrow \quad A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array} \quad \Rightarrow \quad M = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{array}$$

a_{11}

$$\text{Co-factor of } a_{11} = (-1)^{1+1} \begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 1 * \{ 2(6-1) - (-1)(-2-0) + 0(1-0) \}$$

$$= 1 * (10-2) = 8$$

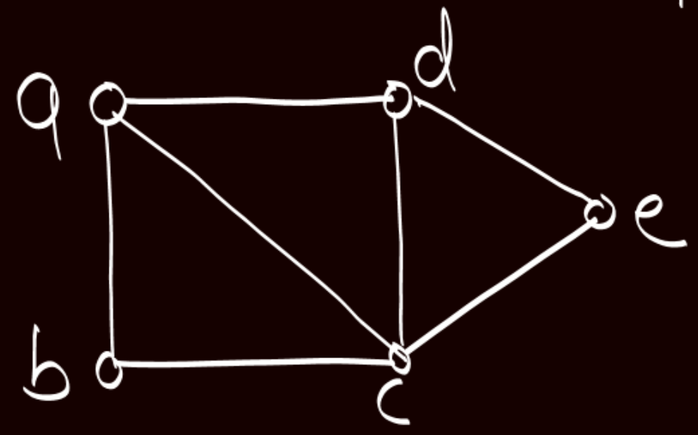
✯ Kirchoff's Matrix Theorem:-

Step 1: Write adjacency matrix 'A' w.r.t. given graph

Step 2: Obtain laplacian matrix 'M' from matrix 'A' by replacing 1's in matrix A by '-1' and by replacing 0's of principle diagonal in matrix 'A' by the degree of corresponding vertex

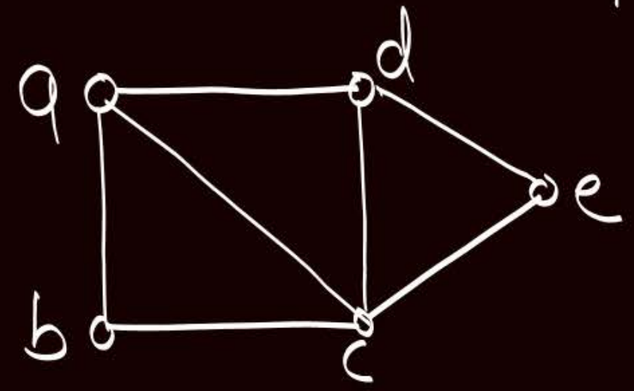
Step 3: Co-factor w.r.t. any element of matrix 'M' will give us the number of spanning tree

Q:- Find the no. of spanning trees of following graph



$$M = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 3 & -1 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ -1 & 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{bmatrix} \end{array}$$

Q:- Find the no. of spanning trees of following graph



$\Rightarrow$ M:

	a	b	c	d	e
a	3	-1	-1	-1	0
b	-1	2	-1	0	0
c	-1	-1	4	-1	-1
d	-1	0	-1	3	-1
e	0	0	-1	-1	2

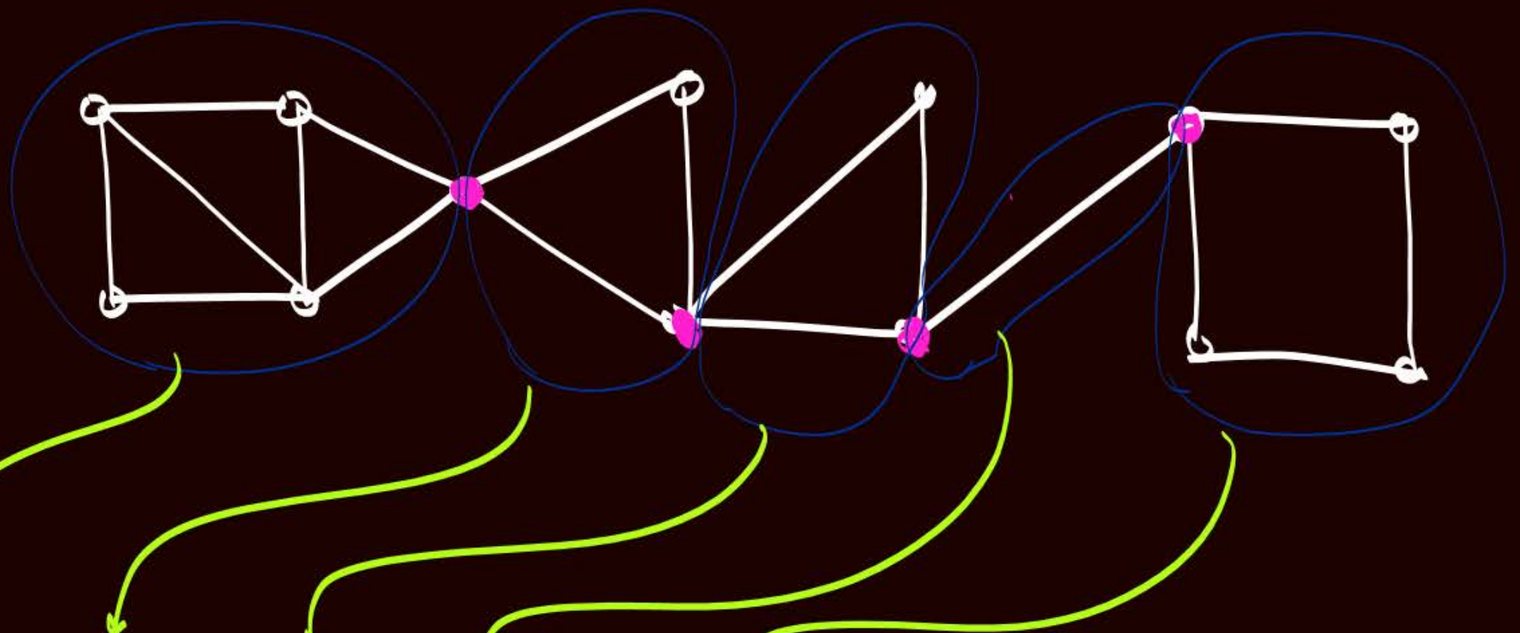
Cofactor of $a_{11} = (-1)^{1+1} \begin{vmatrix} 2 & -1 & 0 & 0 \\ -1 & 4 & -1 & -1 \\ 0 & -1 & 3 & -1 \\ 0 & -1 & -1 & 2 \end{vmatrix} = 2 \times \begin{vmatrix} 4 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 2 \end{vmatrix} - (-1) \times \begin{vmatrix} -1 & -1 & -1 \\ 0 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} + 0 - 0$

$= 2 \times \{(4 \times 5) + 1(-2-1) - 1(1+3)\} + 1\{-1(6-1)\}$

$= 2 \times (20 - 3 - 4) - 5 = 21$ Ans

eg Find the number of spanning trees of the following graph

G =



Total no. of spanning trees = $21 * 3 * 3 * 1 * 4 = \underline{\underline{756}}$



Topic : Caley's Formula

* For a Complete graph K_n number of spanning trees can be given by = $n^{(n-2)}$

Gate-2024: How many spanning trees are possible for a Complete graph with '4' vertices {i.e. K_4}

Ans = $(4)^{4-2} = 4^2 = 16$

Ans



2 mins Summary

Topic Connectivity

Topic Spanning tree

[redacted]

Slide

THANK - YOU